

Stochastic optimization

Two-stage stochastic programming with recourse

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Setting and notation

Uncertainty is represented by **random elements** on a probability space $(\Omega, \mathcal{F}, P)$: a realization ω is drawn from the sample space Ω , and an **event** is a member of the σ -algebra $\mathcal{F}$.

Throughout, ξ denotes a random vector, ξ one of its realizations, and Ξ its support — the smallest closed set with $P[\xi \in \Xi] = 1$.

The **probability background notes** cover the whole setting used here, and in particular the four points this deck leans on without redefining them: the measurability that makes $\mathcal{Q}(x)$ well defined, the closedness of the support, the meaning of “finite second-order moments”, and the difference between “almost surely” and “always”.

A random linear program

Consider the linear program (LP), parametrized by the random vector (r.v.) $\xi : \Omega \rightarrow \mathbb{R}^r$:

$$\begin{aligned} \min_x \quad & c^T x \\ \text{s.t.} \quad & Ax = b \\ & T(\xi)x = h(\xi) \\ & x \in X, \end{aligned}$$

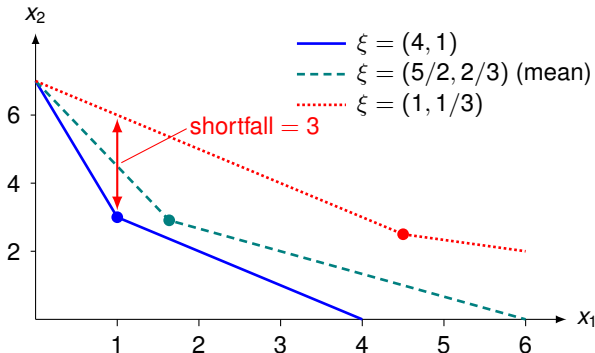
with $X = \{x \in \mathbb{R}^n | l \leq x \leq u\}$. Example:

$$\begin{aligned} \min_x \quad & x_1 + x_2 \\ \text{s.t.} \quad & \xi_1 x_1 + x_2 \geq 7 \\ & \xi_2 x_1 + x_2 \geq 4 \\ & x_1, x_2 \geq 0, \end{aligned}$$

where $\xi_1 \sim U[1, 4]$, $\xi_2 \sim U[1/3, 1]$. Slack variables bring the two inequalities back to the equality form above, with $X = \mathbb{R}_+^2$.

The randomness, seen in the decision space

Each realization tilts the two constraints: the frontier of the feasible set, and with it the best x , moves.

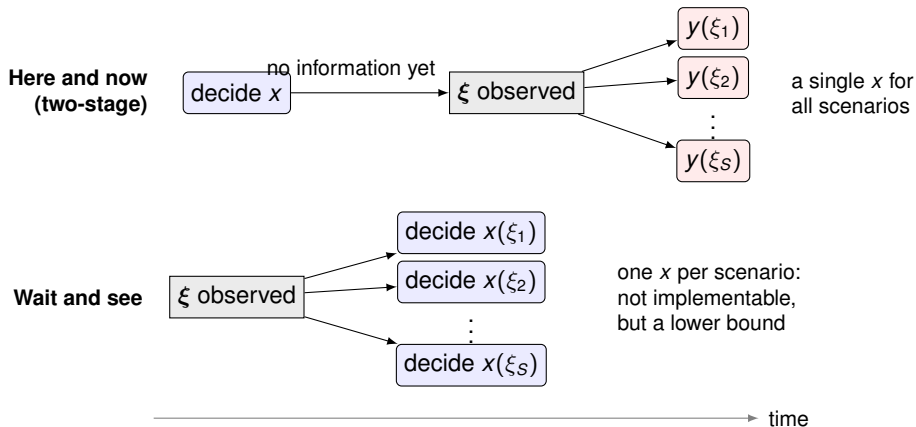


The three dots are the three optimal x : the one chosen for $\xi = (4, 1)$ misses the first constraint by 3 units when $\xi = (1, 1/3)$ occurs.

What to do?

- How to solve this problem?
- What is the *meaning* of solving this problem?
- Can we decide on x *after* having observed the realization of the r.v. ξ ?
We then talk of a **wait-and-see** approach. The problem is then easier to solve (we have here a simple linear program).
- But this approach is rarely appropriate!!! It usually fails to model the real behaviour of the decision process: we have to decide on x **before** we know the realizations of ξ .
- Three suggestions:
 1. try to estimate, predict, the uncertainty;
 2. chance constraints;
 3. penalties on deviations.

When is the decision taken?



Remove the randomness?

A popular approach consists in looking for reasonable values for ξ_1 and ξ_2 . How?

Propositions:

- unbiased: choose the mean values for each random variable;
- pessimistic: choose the worst-case values for ξ ;
- optimistic: choose the best-case values for ξ .

Each approach will deliver a different optimal solution!

Penalization of violations

Again, we have to deal with decision problems where the decision x has to be taken before we know the realization of ξ

In the simplest case, we can simply penalize the constraints deviations by vectors of penalty coefficients q_+ and q_- .

$$\begin{aligned} \min \quad & c^T x + q_+^T s(\xi) + q_-^T t(\xi) \\ \text{s.t.} \quad & Ax = b, \\ & T(\xi)x + s(\xi) - t(\xi) = h(\xi), \\ & x \in X, \quad s(\xi) \geq 0, \quad t(\xi) \geq 0. \end{aligned}$$

But it is still not possible to solve the problem!

The new optimization problem

A reasonable, and solvable, problem is then

$$\begin{aligned} \min \quad & c^T x + \mathbb{E}_\xi [q_+^T s(\xi) + q_-^T t(\xi)] \\ \text{s.t.} \quad & Ax = b, \\ & T(\xi)x + s(\xi) - t(\xi) = h(\xi), \text{ for a.e. } \xi \\ & x \in X, \quad s(\xi) \geq 0, \quad t(\xi) \geq 0, \text{ for a.e. } \xi, \end{aligned}$$

where *a.e.* stands for “almost every”.

- In general, we can react in a correct (and maybe optimal) way: we have a recourse to “correct” the first decision once the uncertainty is removed.
- An LP recourse structure is provided by 3 elements:
 - a set $Y \subset \mathbb{R}^p$ that describes the feasible set of recourse actions, for instance $Y = \{y \in \mathbb{R}^p \mid y \geq 0\}$;
 - q : a vector of recourse costs;
 - $W \in \mathbb{R}^{m \times p}$: **recourse matrix**.

Recourse formulation

More generally, consider the following program:

$$\begin{aligned} \min \quad & c^T x + \mathbb{E}_{\xi}[q(\xi)^T y(\xi)] \\ \text{s.t.} \quad & Ax = b, \\ & T(\xi)x + Wy(\xi) = h(\xi), \text{ for a.e. } \xi \\ & x \in X, \\ & y(\xi) \in Y, \text{ for a.e. } \xi. \end{aligned}$$

We could have W varying with the realization ξ . If W is unique, as in the previous formulation, we speak of **fixed recourse**: the recourse does not change with the scenario.

But how to decide on y ?

The two-stage linear stochastic problem (SP)

We can also rewrite the stochastic programming problem with recourse in terms of x only:

$$\min_{x \in X} \{c^T x + Q(x) \mid Ax = b\}.$$

It is a (nonlinear) mathematical programming problem in $\mathbb{R}^n$.
The properties of $Q(x)$ influence the solution techniques.

Is $Q(x)$

- linear?
- convex?
- continuous?
- differentiable?

Expression in terms of y 's

$$\min_{x, y(\xi)} \mathbb{E}_{\xi}[c^T x + q(\xi)^T y(\xi)]$$

$$\text{s.t. } Ax = b$$

first-stage constraints

$$T(\xi)x + Wy(\xi) = h(\xi), \text{ for a.e. } \xi$$

second-stage constraints

$$x \in X, y(\xi) \in Y \text{ for a.e. } \xi.$$

Consider the (discrete) case where $\Omega = \{\omega_1, \omega_2, \dots, \omega_S\}$, so that $\Xi = \xi(\Omega) \subseteq \mathbb{R}^r$ — the range coincides with the support here, every scenario having a positive probability, but not in general.

$$P[\{\omega_s\}] = p_s, \quad s = 1, 2, \dots, S$$

$$q_s = q(\xi(\omega_s)), \quad T_s = T(\xi(\omega_s)), \quad h_s = h(\xi(\omega_s))$$

Deterministic equivalent

Develop along the S scenarios.

$$\min_{x, y_1, \dots, y_S} c^T x + p_1 q_1^T y_1 + p_2 q_2^T y_2 + \dots + p_S q_S^T y_S$$

s.t.

$$\begin{array}{rcl} Ax & & = b \\ T_1 x & + Wy_1 & = h_1 \\ T_2 x & & + Wy_2 = h_2 \\ \vdots & & \ddots \\ T_S x & & + Wy_S = h_S \end{array}$$

$$x \in X, y_1 \in Y, y_2 \in Y, \dots, y_S \in Y.$$

Deterministic equivalent (cont'd)

- $y_s = y(\xi(\omega_s))$: recourse action to take if the scenario s occurs.
- **Advantage**: linear program.
- **Drawback**: linear program of (very) large dimension:
 - $n + pS$ variables;
 - $m_1 + mS$ constraints, with $A \in \mathbb{R}^{m_1 \times n}$ and $W \in \mathbb{R}^{m \times p}$.
- **Advantage**: the linear program matrix has a special structure (stairway shape).
Can we exploit it?

The staircase structure

$$\begin{array}{cccccc}
 & x & y_1 & y_2 & \cdots & y_S \\
 \left[\begin{array}{c} A \\ T_1 \\ T_2 \\ \vdots \\ T_S \end{array} \right] & & \begin{array}{c} \\ W \\ \\ \\ W \end{array} & & & \\
 & & & W & & \\
 & & & & \ddots & \\
 & & & & & W
 \end{array} = \begin{bmatrix} b \\ h_1 \\ h_2 \\ \vdots \\ h_S \end{bmatrix}$$

the only coupling between scenarios:
the first-stage decision x

Delete that first column and the problem falls apart into S independent small programs: this is what the decomposition methods of the next chapter exploit.

Large scale,... and?

Assume that we have r random variables ($\Xi \subseteq \mathbb{R}^r$).

- Consider the following problem (source: Linderoth). A Telecom company wants to expand its network in order to meet an unknown (random) demand.
- There are 86 unknown demands. Each demand is independent and takes a value in a set of 7 values. Consequently

$$S = |\Omega| = 7^{86} \approx 4.77 \times 10^{72}.$$

For comparison, the observable universe is estimated to hold some 10^{80} atoms.

- It can be even worse. . .
If Ω is not finite, but holds an infinite number of elements? It is especially true with continuous random variables. Our “deterministic equivalent” would have an infinite number of variables and constraints!
- We can solve an approximate problem, obtained by sampling over the random vector.

An example (cont'd)

Consider again our toy problem

$$\begin{aligned} \min_x \quad & x_1 + x_2 \\ \text{s.t.} \quad & \xi_1 x_1 + x_2 \geq 7 \\ & \xi_2 x_1 + x_2 \geq 4 \\ & x_1, x_2 \geq 0, \end{aligned}$$

where $\xi_1 \sim U[1, 4]$, $\xi_2 \sim U[1/3, 1]$.

How to build the deterministic equivalent?

Example: recourse formulation

Assume for now that Ω is finite, with S scenarios.

$$\begin{aligned} \min_{x,y} \quad & x_1 + x_2 + \sum_{s=1}^S p_s q(y_{1s} + y_{2s}) \\ \text{s.t.} \quad & \xi_{1s}x_1 + x_2 + y_{1s} \geq 7, & s = 1, \dots, S \\ & \xi_{2s}x_1 + x_2 + y_{2s} \geq 4, & s = 1, \dots, S \\ & x_1, x_2 \geq 0, \\ & y_{1s}, y_{2s} \geq 0, & s = 1, \dots, S. \end{aligned}$$

A difficulty is therefore to decide how to construct the deterministic equivalent.
How to choose q ?

How to construct the scenarios? We can proceed with Monte Carlo sampling, with $p_s = 1/S, \forall s$. We will explore this approach in more details later.

Example: recourse formulation (cont'd)

More generally, we can build the program

$$\begin{aligned} \min_x \quad & x_1 + x_2 + \mathbb{E}_\xi[Q(x, \xi)] \\ \text{s.t.} \quad & x_1, x_2 \geq 0, \end{aligned}$$

and

$$\begin{aligned} Q(x, \xi) = \min_y \quad & q_1 y_1 + q_2 y_2 \quad (q_1 = q_2 = q \text{ above}) \\ \text{s.t.} \quad & \xi_1 x_1 + x_2 + y_1 \geq 7, \\ & \xi_2 x_1 + x_2 + y_2 \geq 4, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Two-stage linear programming problem, fixed recourse

We go a bit further by considering the problem

$$\min c^T x + \mathbb{E}_{\xi}[q(\xi)^T y(\xi)]$$

subject to the constraints

$$\begin{aligned} Ax &= b, \\ T(\xi)x + Wy(\xi) &= h(\xi) \quad \text{for a.e. } \xi \in \Xi, \\ x &\in X, \\ y(\xi) &\in Y \quad \text{for a.e. } \xi \in \Xi, \end{aligned}$$

where ξ is a random vector defined on the probability space $(\Omega, \mathcal{F}, P)$, and Ξ is the support of ξ .

Unless stated otherwise, ξ **gathers the random entries** of q , h and T , as in Birge and Louveaux. This loses no generality, and the hypotheses made on ξ below — moments, absolute continuity — are then hypotheses on the data themselves.

Reformulation(s)

The problem becomes

$$\min_{x \in X | Ax=b} \left\{ c^T x + \mathbb{E}_{\xi} \left[\min_{y \in Y} \left\{ q(\xi)^T y \mid Wy = h(\xi) - T(\xi)x \right\} \right] \right\},$$

or

$$\min_{x \in X | Ax=b} c^T x + \mathbb{E}_{\xi}[Q(x, \xi)],$$

with

$$Q(x, \xi) = \min_{y \in Y} \left\{ q(\xi)^T y : Wy = h(\xi) - T(\xi)x \right\}.$$

Second-stage function, or **recourse function**, $v : \Xi \times \mathbb{R}^m \rightarrow \overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}$:

$$v(\xi, z) \stackrel{\text{def}}{=} \min_{y \in Y} \{ q(\xi)^T y \mid Wy = z \}.$$

Given a “policy” x and a realization ξ , z measures the deviation of the first stage, i.e. $z = h(\xi) - T(\xi)x$, $v(\xi, z)$ is the minimum cost to “correct” the decision.

Recourse function

Expected recourse function:

$$Q(x) \stackrel{\text{def}}{=} \mathbb{E}_{\xi}[Q(x, \xi)],$$

with

$$Q(x, \xi) = v(\xi, h(\xi) - T(\xi)x).$$

The problem can now be rewritten as:

$$\min_{x \in X} c^T x + Q(x) \text{ such that } Ax = b.$$

It is a nonlinear program over $\mathbb{R}^n$. Properties?

Formulations summary

Set $X = \mathbb{R}_+^n$.

$$\min_{x \in \mathbb{R}_+^n \mid Ax=b} \left\{ c^T x + \mathbb{E}_\xi \left[\min_{y \in \mathbb{R}_+^p} \{ q(\xi)^T y \mid Wy = h(\xi) - T(\xi)x \} \right] \right\}$$

$$\min_{x \in \mathbb{R}_+^n \mid Ax=b} \left\{ c^T x + \mathbb{E}_\xi [v(\xi, h(\xi) - T(\xi)x)] \right\}$$

$$\min_{x \in \mathbb{R}_+^n \mid Ax=b} \left\{ c^T x + \mathbb{E}_\xi [Q(x, \xi)] \right\}$$

$$\min_{x \in \mathbb{R}_+^n} \left\{ c^T x + Q(x) \mid Ax = b \right\}$$

Notations

- First-stage feasible set:

$$K_1 = \{x \in \mathbb{R}_+^n \mid Ax = b\}.$$

- Second-stage strong feasible set:

$$K_2^s = \{x \mid Q(x) < \infty\}.$$

Therefore we can rewrite the problem as

$$\min_x \{c^T x + Q(x) \mid x \in K_1 \cap K_2^s\}.$$

Weak feasible set

See Walkup, D. W. and Wets, R. J.-B., *Stochastic programs with recourse*, SIAM Journal on Applied Mathematics, 15(5):1299–1314, 1967.

Positive hull (or conical hull)

$$\text{pos } W = \{z \mid z = Wy, y \in \mathbb{R}_+^p\}$$

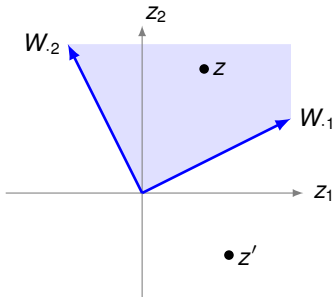
(positive hulls and Farkas' lemma are recalled in the linear programming background notes)

Weak second-stage feasible set

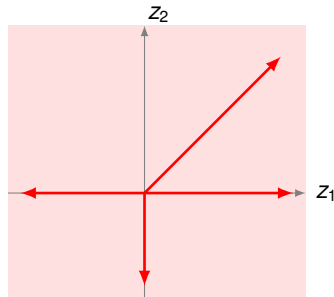
$$\begin{aligned} K_2 &= \{x \in \mathbb{R}^n \mid Q(x, \xi) < +\infty \text{ a.s.}\} \\ &= \{x \in \mathbb{R}^n \mid (h(\xi) - T(\xi)x) \in \text{pos } W \text{ a.s.}\} \end{aligned}$$

pos W , graphically

pos W is the cone spanned by the columns of W with non-negative weights: the set of right-hand sides z that the recourse can absorb.



$z \in \text{pos } W: Q(x, \xi) \text{ finite}$
 $z' \notin \text{pos } W: Q(x, \xi) = +\infty$



pos $W = \mathbb{R}^2$: **complete recourse**,
every z is absorbed

Relatively complete recourse

A problem is said to have a **relatively complete recourse** if $K_1 \subseteq K_2$.

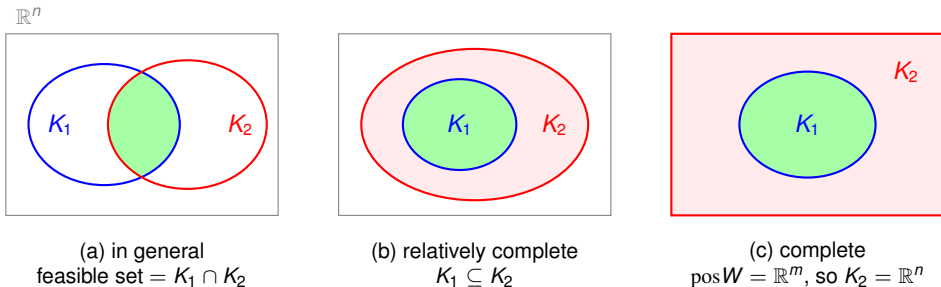
Advantage: the second-stage problem is feasible $\forall x$ feasible in the first stage, almost surely (i.e. with probability equal to 1).

Issue: $K_2^s \subseteq K_2$. We would like $K_2^s = K_2$.

Complete recourse

- The relatively complete recourse is very useful in practice and on a theoretical point of view, but it can be difficult to identify.
- **Complete recourse**: $\text{pos } W = \mathbb{R}^m$, i.e. $\forall z \in \mathbb{R}^m, \exists y \in \mathbb{R}_+^p$ such that $Wy = z$.
- Complete recourse $\Rightarrow$ relatively complete recourse, as $\forall x, T(\xi), h(\xi), Q(x, \xi) < \infty$ with $z = h(\xi) - T(\xi)x$.

K_1 , K_2 , and the two notions of completeness



Completeness is read off W alone; relative completeness also depends on A , b and on the support of ξ , and is the property that really matters: it is what forbids a first-stage decision to be caught infeasible in the second stage.

$$K_2 \neq K_2^s$$

Consider the second-stage problem

$$\begin{aligned} \min_y \quad & 2y_1 + y_2 \\ \text{s.t.} \quad & y_1 + y_2 \geq 1 - x_1, \\ & y_1 \geq \xi - x_1 - x_2, \\ & y_1, y_2 \geq 0, \end{aligned}$$

with

$$P[\xi = 2^n] = \frac{1}{2^{n+1}}, \quad n = 0, 1, 2, \dots$$

Note that $P[\xi = 2^n] \in (0, 1)$, $n = 0, 1, 2, \dots$, $\Xi = \{2^n, n = 0, 1, 2, \dots\}$, and

$$\sum_{n=0}^{+\infty} P[\xi = 2^n] = \sum_{n=0}^{+\infty} 2^{-n-1} = \frac{1}{2} \frac{1}{1 - \frac{1}{2}} = 1.$$

$K_2 \neq K_2^s$ (cont'd)

Given $x \in \mathbb{R}^2$, we have

$$\begin{aligned} Q(x) &= \mathbb{E}_{\xi} \left[\min_{y \geq 0} 2y_1 + y_2 \mid y_1 + y_2 \geq 1 - x_1, y_1 \geq \xi - x_1 - x_2 \right] \\ &\geq \mathbb{E}_{\xi} \left[\min_{y_1 \geq 0} 2y_1 \mid y_1 \geq \xi - x_1 - x_2 \right] \\ &= 2\mathbb{E}_{\xi} [\max\{0, \xi - x_1 - x_2\}] \\ &= 2 \sum_{\xi \in \Xi} P[\xi = \xi] \max\{0, \xi - x_1 - x_2\} \\ &= 2 \sum_{n=0}^{+\infty} 2^{-n-1} \max\{0, 2^n - x_1 - x_2\} \\ &= \sum_{n=0}^{+\infty} 2^{-n} \max\{0, 2^n - x_1 - x_2\} = +\infty \end{aligned}$$

$$K_2 \neq K_2^s \text{ (cont'd)}$$

Thus, $K_2^s = \emptyset$.

The problem, set under standard form, can be rewritten as

$$\begin{aligned} \min_{y,u} \quad & 2y_1 + y_2 \\ \text{s.t.} \quad & y_1 + y_2 - u_1 = 1 - x_1, \\ & y_1 - u_2 = \xi - x_1 - x_2, \\ & y_1, y_2, u_1, u_2 \geq 0, \end{aligned}$$

and

$$W = \begin{pmatrix} 1 & 1 & -1 & 0 \\ 1 & 0 & 0 & -1 \end{pmatrix}.$$

pos $W = ?$

$$K_2 \neq K_2^s \text{ (cont'd)}$$

Complete recourse if $\text{pos } W = \mathbb{R}^2$, i.e.

$$\forall z \in \mathbb{R}^2, \exists y \geq 0, u \geq 0 \text{ s.t. } W \begin{pmatrix} y \\ u \end{pmatrix} = z.$$

We have that

$$\begin{aligned} W \begin{pmatrix} y \\ u \end{pmatrix} &= \begin{pmatrix} y_1 + y_2 - u_1 \\ y_1 - u_2 \end{pmatrix} = z \\ \Leftrightarrow \begin{cases} y_1 + y_2 - u_1 &= z_1, \\ y_1 - u_2 &= z_2. \end{cases} \end{aligned}$$

Thus, we can take

$$\begin{cases} y_1 = z_2, u_2 = 0, & \text{if } z_2 \geq 0, \\ y_1 = 0, u_2 = -z_2, & \text{otherwise.} \end{cases}$$

$$K_2 \neq K_2^S \text{ (cont'd)}$$

Then, writing $y_2 - u_1 = z_1 - y_1$, we take

$$\begin{cases} y_2 = z_1 - y_1, & u_1 = 0, & \text{if } z_1 - y_1 \geq 0, \\ y_2 = 0, & u_1 = y_1 - z_1, & \text{otherwise.} \end{cases}$$

Thus,

- $\text{pos} W = \mathbb{R}^2$, and the recourse is complete,
- $K_2 = \mathbb{R}^2$, and the recourse is relatively complete,
- $K_2 \neq K_2^S$.

Note that

$$\mathbb{E}[\xi] = \sum_{n=0}^{+\infty} P[\xi = 2^n] 2^n = \sum_{n=0}^{+\infty} \frac{2^n}{2^{n+1}} = \sum_{n=0}^{+\infty} \frac{1}{2} = +\infty.$$

$$K_2 \neq K_2^s \text{ (cont'd)}$$

$\mathbb{E}[\xi]$ not finite does not necessarily imply that $Q(x) = +\infty$. Consider

$$\begin{aligned} \min_y \quad & y_1 - y_2 \\ \text{s.t.} \quad & y_1 \geq \xi, \\ & y_2 \leq \xi, \\ & y_1, y_2 \geq 0. \end{aligned}$$

The optimum is attained at $y_1 = y_2 = \xi$, so $Q(x, \xi) = 0$ for all ξ (and all x), and therefore $Q(x) = 0$.

$$K_2 = K_2^s$$

Theorem

For a stochastic program with fixed recourse, where ξ has finite second-order moments ($\mathbb{E}[\|\xi\|^2] < \infty$; see the probability notes for why this is the hypothesis that recurs),

$$P[Q(x, \xi) < \infty] = 1 \implies Q(x) < \infty,$$

and consequently

$$K_2 = K_2^s.$$

Elementary feasible set

- Given a realization ξ , the **elementary second-stage feasible set** is defined as:

$$K_2(\xi) = \{x \mid Q(x, \xi) < \infty\}.$$

- Define **possibility interpretation** of the second-stage feasibility as

$$K_2^P = \cap_{\xi \in \Xi} K_2(\xi).$$

- Clearly $K_2 = K_2^P$ if ξ has a finite support, every scenario having a positive probability. Is it still the case when ξ follows a continuous distribution?

Elementary feasible set (cont'd)

Theorem

For a stochastic program with fixed recourse, where ξ has finite second-order moments,

$$K_2 = K_2^S = K_2^P.$$

With a fixed recourse, $x \in K_2(\xi) \iff h(\xi) - T(\xi)x \in \text{pos} W$, a **closed** cone: the set of favourable ξ is then closed, and having probability one forces it to contain the whole support. The exercise below shows what happens when the recourse is not fixed.

K_2 is a polyhedron

Theorem

For a stochastic program with fixed recourse where ξ has a *finite support*, K_2 is a polyhedron.

$\text{pos}W$ is a polyhedral cone, so it is the intersection of finitely many half-spaces through the origin: writing $\sigma^1, \dots, \sigma^r$ for the generators of its *polar cone* (background notes),

$$z \in \text{pos}W \iff (\sigma^k)^T z \leq 0, \quad k = 1, \dots, r.$$

With a fixed recourse, $x \in K_2(\xi) \iff h(\xi) - T(\xi)x \in \text{pos}W$, hence

$$K_2(\xi) = \{x \mid (\sigma^k)^T (h(\xi) - T(\xi)x) \leq 0, \quad k = 1, \dots, r\},$$

a polyhedron. When $\Xi = \{\xi_1, \dots, \xi_S\}$ is finite, $K_2 = \bigcap_{s=1}^S K_2(\xi_s)$ is a finite intersection of polyhedra, hence a polyhedron.

Those inequalities are exactly the *feasibility cuts* of the next chapter, which never enumerates the σ^k but discovers, one at a time, the few that the current iterate violates. On an infinite support the intersection is no longer finite and K_2 , while still closed and convex, need not be polyhedral.

Simple recourse

A particular case of complete recourse is the **simple recourse**, for which we have

$$W = \begin{pmatrix} I & -I \end{pmatrix},$$

with I the identity matrix, of order m .

pos $W = \mathbb{R}^m$: writing

$$y = \begin{pmatrix} y^+ \\ y^- \end{pmatrix},$$

where $y \in \mathbb{R}^{2m}$, $y^+, y^- \in \mathbb{R}^m$, for $z \in \mathbb{R}^m$, we can solve

$$Wy = z, \quad y \geq 0,$$

by choosing $y^+ = \max\{z, 0\}$, $y^- = \max\{-z, 0\}$.

Simple recourse (cont'd)

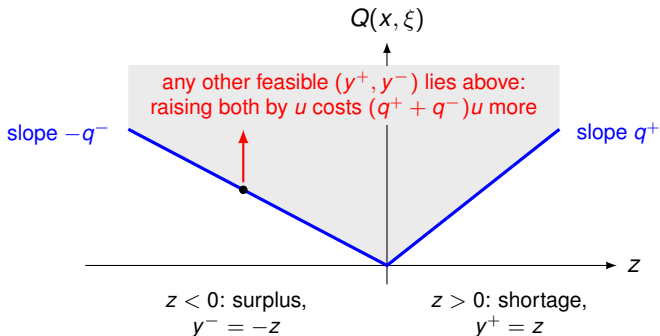
The second stage program can be read as

$$\begin{aligned} Q(x, \xi) = \min_y & q^+(\xi)^T y^+ + q^-(\xi)^T y^- \\ \text{s.t. } & y^+ - y^- = h(\xi) - T(\xi)x, \\ & y^+, y^- \geq 0. \end{aligned}$$

That is, for $q^+(\xi) + q^-(\xi) \geq 0$, at most one of y_i^+ and y_i^- is positive at the optimum, so they measure the positive and negative parts of the discrepancy $h(\xi) - T(\xi)x$.

Simple recourse, graphically

With $m = 1$, write $z = h(\xi) - T(\xi)x$ for the discrepancy left by the first stage.



The V points upwards exactly when $q^+ + q^- \geq 0$; otherwise inflating y^+ and y^- together is profitable and $Q(x, \xi) = -\infty$.

Simple recourse (cont'd)

- Since the recourse is complete, it is feasible for any first-stage decision x .
- The following result establishes conditions to have a finite recourse (i.e. $|Q(x)| < \infty$).

Theorem

Assume that the two-stage (linear) stochastic program is feasible and has a simple recourse, and that ξ has finite second-order moments. Then $Q(x)$ is finite if and only if, for any component i , $\mathbf{q}_i^+ + \mathbf{q}_i^- \geq 0$ with probability one.

Simple recourse (cont'd)

Proof.

($\Rightarrow$) Assume by contradiction that $\mathcal{Q}$ is finite, but for some component i , $q_i^+(\xi(\omega)) + q_i^-(\xi(\omega)) < 0$ for $\omega \in \Omega_1$ with $P[\Omega_1] > 0$. Then, for any feasible x , for all $\omega \in \Omega_1$ with $h_i(\xi(\omega)) - T_i(\xi(\omega))x > 0$, define

$$y_i^+(\xi(\omega)) = h_i(\xi(\omega)) - T_i(\xi(\omega))x + u, \quad y_i^-(\xi(\omega)) = u.$$

Therefore,

$$y_i^+(\xi(\omega)) - y_i^-(\xi(\omega)) = h_i(\xi(\omega)) - T_i(\xi(\omega))x, \quad y_i^+ \geq 0, \quad y_i^- \geq 0.$$

Moreover, we can choose y_j^+ and y_j^- feasible, $j \neq i$.



Simple recourse (cont'd)

Proof.

($\Rightarrow$)

When $u \rightarrow \infty$, $Q(x, \xi(\omega)) \rightarrow -\infty$ since $q_i^+(\xi(\omega))y_i^+ + q_i^-(\xi(\omega))y_i^- \rightarrow -\infty$.

A similar argument can be applied if $h_i(\xi(\omega)) - T_i(\xi(\omega))x \leq 0$, by taking

$$y_i^+(\xi(\omega)) = u, \quad y_i^-(\xi(\omega)) = -h_i(\xi(\omega)) + T_i(\xi(\omega))x + u.$$

The two cases cover every $\omega \in \Omega_1$, so $Q(x, \xi(\omega)) = -\infty$ on a set of positive probability and $Q(x) = -\infty$: the recourse is not finite. □

Simple recourse (cont'd)

Proof.

($\Leftarrow$) Assume $\forall i, P[\mathbf{q}_i^+ + \mathbf{q}_i^- \geq 0] = 1$. Write $z(\xi) = h(\xi) - T(\xi)x$. Any feasible (y^+, y^-) satisfies $y^+ - y^- = z(\xi)$, $y^+ \geq 0$, $y^- \geq 0$, so $y_i^- = y_i^+ - z_i(\xi)$ and its **objective value** is

$$\begin{aligned} \sum_{i=1}^m (q_i^+(\xi)y_i^+ + q_i^-(\xi)y_i^-) &= \sum_{i=1}^m (q_i^+(\xi)y_i^+ + q_i^-(\xi)(y_i^+ - z_i(\xi))) \\ &= \sum_{i=1}^m ((q_i^+(\xi) + q_i^-(\xi))y_i^+ - q_i^-(\xi)z_i(\xi)). \end{aligned}$$

This is the value at an arbitrary feasible point, not yet the optimal value $Q(x, \xi)$.



Simple recourse (cont'd)

Proof.

($\Leftarrow$)

Almost surely $q_i^+(\xi) + q_i^-(\xi) \geq 0$, and $y_i^+ \geq 0$, so the first sum above is non-negative: **every** feasible point has an objective value at least $-q^-(\xi)^T z(\xi)$. Taking the infimum,

$$Q(x, \xi) \geq -q^-(\xi)^T z(\xi) \quad \text{a.s.}$$

The program is feasible and bounded below, so it admits an optimal solution.

The objective being non-decreasing in each y_i^+ , the infimum is attained at the smallest feasible y_i^+ (equivalently, at a basic solution, where at most one of y_i^+ , y_i^- is positive):

$$y^+(\xi) = (h(\xi) - T(\xi)x)^+, \quad y^-(\xi) = (-h(\xi) + T(\xi)x)^+,$$

where $a^+ = \max\{0, a\}$.



Simple recourse (cont'd)

Proof.

($\Leftarrow$)

At that optimum $Q(x, \xi) = \sum_{i=1}^m (q_i^+(\xi) z_i(\xi)^+ + q_i^-(\xi) (-z_i(\xi))^+)$, hence $|Q(x, \xi)| \leq \|q(\xi)\| \|z(\xi)\|$.

This is where the finite second-order moments are used: ξ gathers the random entries of q , h and T (the convention stated with the fixed-recourse problem), and $z(\xi) = h(\xi) - T(\xi)x$ is linear in (h, T) , so $\mathbb{E}[\|q(\xi)\|^2]$ and $\mathbb{E}[\|z(\xi)\|^2]$ are finite, and Cauchy–Schwarz gives

$$\mathbb{E}[\|q(\xi)\| \|z(\xi)\|] \leq \left(\mathbb{E}[\|q(\xi)\|^2] \right)^{1/2} \left(\mathbb{E}[\|z(\xi)\|^2] \right)^{1/2} < \infty.$$

Therefore $|Q(x)| < \infty$: the recourse is finite, and in particular $Q(x) \geq \mathbb{E}_{\xi} [-q^-(\xi)^T z(\xi)]$.



Exercise

Consider the second stage program

$$Q(x, \xi) = \min_y \{y \mid \xi y = 1 - x, y \geq 0\}.$$

We assume that ξ follows a triangular distribution on $[0, 1]$, with $P[\xi \leq u] = u^2$.

(a) Is the recourse fixed? Why?

The recourse is not fixed, as $W \equiv \xi$, and therefore, W is random. Moreover, as ξ can take the value 0, the transformation

$$y = 1/\xi - x/\xi,$$

is not properly defined on $\Xi = [0, 1]$; this also means that

$$\text{pos } W = \begin{cases} \{0\} & \text{if } \xi = 0; \\ \mathbb{R}_+ & \text{if } \xi \neq 0. \end{cases}$$

Exercise (cont'd)

(b) Express $K_2(\xi)$ for all ξ in $[0, 1]$.

We have to consider two cases: $\xi = 0$ or $\xi \in (0, 1]$.

1. $\xi \in (0, 1]$ In this case, as $y, \xi \geq 0$, $1 - x$ has to be non-negative in order to have a well-defined problem:

$$K_2(\xi) = \{x \mid x \leq 1\}.$$

The value and optimal solutions are

$$Q(x, \xi) = (1 - x)/\xi, \quad y^* = (1 - x)/\xi.$$

2. $\xi = 0$ There exists no y such that $0 \cdot y = 1 - x$, except if $x = 1$, so

$$K_2(0) = \{1\}.$$

Exercise (cont'd)

(c) Express K_2 , K_2^P and Q .

From the previous point, we have

$$K_2^P = \{x \mid x \leq 1\} \cap \{1\} = \{1\}.$$

We also have, as $P[\xi = 0] = 0$,

$$Q(x) = \int_0^1 \frac{1-x}{\xi} 2\xi d\xi = 2(1-x), \forall x \leq 1.$$

Consequently $K_2^P \subset K_2 = \{x \leq 1\}$.

The difference comes from the fact that a point is not in K_2^P as soon as it is not feasible for a given value of ξ , but K_2 does not consider infeasible situations that occur with a null probability. With a **fixed** recourse — not the case here — and finite second-order moments, we would have had $K_2 = K_2^P$.

Convexity of $v(\xi, \cdot)$

Let y_1^* and y_2^* be two optimal solutions of $v(\xi, z)$, associated to $z = z_1$ and $z = z_2$, respectively. Then, the convex combination $y_\alpha \stackrel{\text{def}}{=} \alpha y_1^* + (1 - \alpha)y_2^*$, $\alpha \in [0, 1]$, is feasible with respect to $z_\alpha = \alpha z_1 + (1 - \alpha)z_2$, as $\alpha y_1^* + (1 - \alpha)y_2^* \geq 0$, and

$$W(\alpha y_1^* + (1 - \alpha)y_2^*) = \alpha W y_1^* + (1 - \alpha)W y_2^* = \alpha z_1 + (1 - \alpha)z_2 = z_\alpha.$$

Since y_α is feasible for z_α , the definition of v as an infimum gives directly

$$\begin{aligned} v(\xi, z_\alpha) &\leq q(\xi)^T y_\alpha = q(\xi)^T (\alpha y_1^* + (1 - \alpha)y_2^*) \\ &= \alpha q(\xi)^T y_1^* + (1 - \alpha)q(\xi)^T y_2^* \\ &= \alpha v(\xi, z_1) + (1 - \alpha)v(\xi, z_2). \end{aligned}$$

In other words, v is a convex function w.r.t. $z \in \mathbb{R}^m$. No optimal solution has to be assumed at z_α : only feasibility of y_α is used.

Convexity of second-stage function

$Q(x, \xi)$ is the value function of a linear program, seen as a function of its right-hand side $h(\xi) - T(\xi)x$ and of its costs: the properties below are those recalled in the linear programming background notes.

We have the following result (Birge and Louveaux, Chapter 3, Theorem 5).

Theorem

For a stochastic program with fixed recourse, $Q(x, \xi)$ is

- (a) a piecewise-linear convex function in (h, T) ,*
- (b) a piecewise-linear concave function in q ,*
- (c) a piecewise-linear convex function in x , for all x in $K = K_1 \cap K_2$.*

(a) and (c) follow from the convexity of $v(\xi, \cdot)$ just established, composed with the affine map $x \mapsto h(\xi) - T(\xi)x$; (b) is the concavity of the value function in the costs; the piecewise linearity comes from the finite number of optimal bases.

Convexity of the recourse

$Q(x) = \mathbb{E}_{\xi}[Q(x, \xi)]$. Suppose for now that ξ has a finite support, $\Xi = \{\xi_1, \xi_2, \dots, \xi_S\}$, so that $Q(x) = \sum_{s=1}^S P[\xi = \xi_s] Q(x, \xi_s)$.

Theorem

A non-negative combination of convex functions is convex: if $f_1, \dots, f_K$ are convex and $\alpha_1, \dots, \alpha_K \geq 0$, then $\sum_{k=1}^K \alpha_k f_k$ is convex.

$Q(x)$ is therefore a convex function w.r.t. x .

What is happening in the continuous case? The same argument applies verbatim to any measure: if $g(\cdot, t)$ is convex for every t , then $x \mapsto \int g(x, t) d\mu(t)$ is convex, an integral being a limit of non-negative combinations. Since

$$Q(x) = \int_{\Xi} Q(x, t) dF(t),$$

$Q(x)$ is convex.

An example...

Consider the second-stage function $Q(x, \xi)$ defined as:

$$\min y^+ + y^- \text{ s.t. } y^+ - y^- = \xi - x, y^+ \geq 0, y^- \geq 0.$$

In other words:

$$y = \begin{pmatrix} y^+ \\ y^- \end{pmatrix} \quad q = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, W = \begin{pmatrix} 1 & -1 \end{pmatrix} \quad h(\xi) = \xi \quad T(\xi) = 1.$$

Relying on the fundamental theorem of linear programming, we are looking for an optimal basic solution, implying that $y^+ = 0$ or $y^- = 0$.

An example... (cont'd)

We immediately see that

$$y^+ = \begin{cases} \xi - x & \text{if } \xi - x \geq 0, \\ 0 & \text{otherwise,} \end{cases}$$

and

$$y^- = \begin{cases} -\xi + x & \text{if } \xi - x < 0, \\ 0 & \text{otherwise.} \end{cases}$$

An example: the dual and the optimality conditions

Dual, with the slacks s_1, s_2 made explicit:

$$\begin{aligned} \max_{\pi} (\xi - x)\pi \quad & \text{s.t. } \pi \leq 1, -\pi \leq 1 \\ \iff \max_{\pi} (\xi - x)\pi \quad & \text{s.t. } \pi + s_1 = 1, -\pi + s_2 = 1, s_1, s_2 \geq 0. \end{aligned}$$

The recourse is simple, and the primal-dual/KKT conditions of the background notes give

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \pi + \begin{pmatrix} s_1 \\ s_2 \end{pmatrix} \quad \text{(dual feasibility)}$$

$$y^+ - y^- = \xi - x \quad \text{(primal feasibility)}$$

$$y^+ \geq 0, y^- \geq 0, s_1 \geq 0, s_2 \geq 0$$

$$s_1 y^+ = 0, s_2 y^- = 0 \quad \text{(complementary slackness).}$$

An example (cont'd)

- The first condition implies that we cannot have $s_1 = s_2 = 0$.
- From the complementarity conditions, we have that $y^+ = 0$ or $y^- = 0$.
- We have to consider two cases:
 - $x \leq \xi$: in this situation, we have

$$y^+ = \xi - x, \quad y^- = 0.$$

- $x \geq \xi$: then,

$$y^- = x - \xi, \quad y^+ = 0.$$

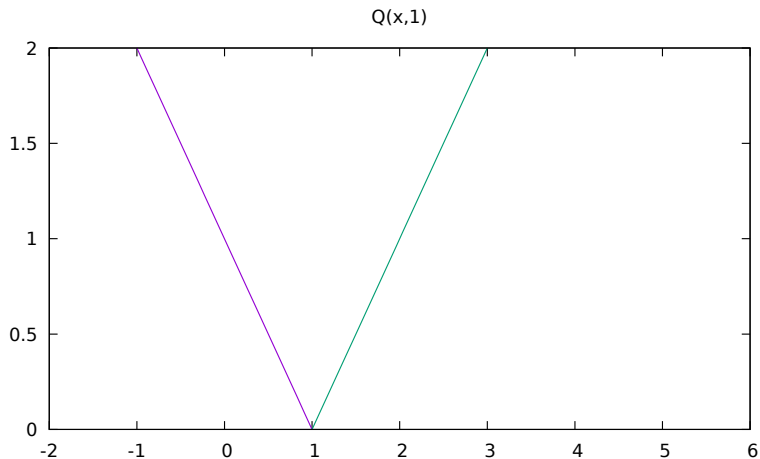
- Consequently,

$$Q(x, \xi) = \begin{cases} \xi - x & \text{if } x \leq \xi, \\ x - \xi & \text{if } x \geq \xi, \end{cases}$$

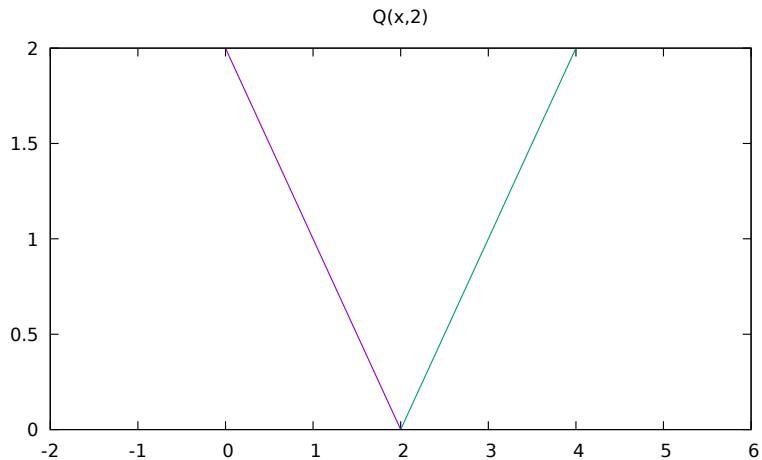
that is, $Q(x, \xi) = |\xi - x|$.

Graphically?

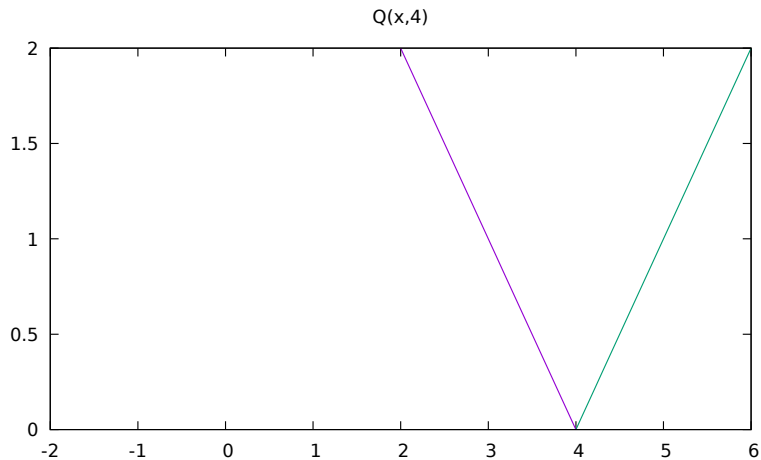
Assume that ξ can take the realizations 1, 2, 4.



Graphically (cont'd)



Graphically (cont'd)



$$Q(x)$$

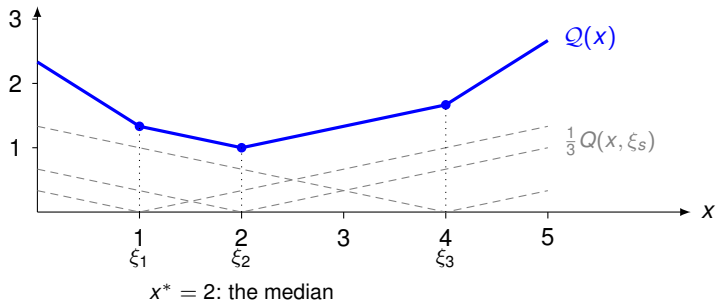
What about $Q(x)$?

Assume that the three realizations have the same probability.

We have to consider 4 cases:

1. $x \leq 1$: $Q(x) = 7/3 - x$;
2. $1 \leq x \leq 2$: $Q(x) = 5/3 - x/3$;
3. $2 \leq x \leq 4$: $Q(x) = x/3 + 1/3$;
4. $4 \leq x$: $Q(x) = x - 7/3$;

Where the kinks come from



Each scenario contributes one kink, at its own realization; Q is their non-negative weighted sum, hence convex and piecewise linear, and its minimizer is here the median of the scenarios.

Properties of $\mathcal{Q}(x)$

$\mathcal{Q}(x)$ is convex; it is moreover piecewise linear **when the support of ξ is finite**, where it is a non-negative weighted sum of the $Q(\cdot, \xi_s)$, each of which has just been shown to be convex and piecewise linear. On a continuous support the piecewise linearity is lost, as the smoothing frame below shows. The general statement is the following.

Theorem

For a stochastic program with fixed recourse where ξ has finite second-order moments,

- (a) $\mathcal{Q}(x)$ is convex, and finite and Lipschitz continuous on K_2 ;
- (b) *when ξ has a finite support, $\mathcal{Q}(x)$ is piecewise linear.*

Reminder: a function f is Lipschitz if there exists some $M < \infty$ such that for all x, y ,

$$|f(x) - f(y)| \leq M\|x - y\|.$$

Differentiability of the recourse

Is $Q(x)$ also differentiable?

The recourse function is partially differentiable with respect to x_j at $(\hat{x}, \hat{\xi})$ if the directional derivative exists for the direction e_j . In other words, there exists a function $\frac{\partial Q(x, \xi)}{\partial x_j}$ such that

$$\frac{Q(\hat{x} + he_j, \hat{\xi}) - Q(\hat{x}, \hat{\xi})}{h} = \frac{\partial Q(\hat{x}, \hat{\xi})}{\partial x_j} + \frac{\rho_j(\hat{x}, \hat{\xi}; h)}{h},$$

with

$$\frac{\rho_j(\hat{x}, \hat{\xi}; h)}{h} \rightarrow 0, \text{ as } h \rightarrow 0.$$

We will assume from now that $\nabla_x Q(x, \xi) = \left(\frac{\partial Q(x, \xi)}{\partial x_1}, \dots, \frac{\partial Q(x, \xi)}{\partial x_n} \right)$ exists.

Differentiability of the recourse (cont'd)

What about the differentiability of $Q(x)$?

$$\begin{aligned}\frac{Q(\hat{x} + he_j) - Q(\hat{x})}{h} &= \int_{\Xi} \frac{Q(\hat{x} + he_j, \xi) - Q(\hat{x}, \xi)}{h} dP \\ &= \int_{\Xi \setminus N_\delta} \frac{\partial Q(\hat{x}, \xi)}{\partial x_j} dP + \int_{\Xi \setminus N_\delta} \frac{\rho_j(\hat{x}, \xi; h)}{h} dP,\end{aligned}$$

where N_δ is measurable and $P[N_\delta] = 0$. Therefore, we have

Theorem

If $Q(x, \xi)$ is partially differentiable almost everywhere, if its partial derivative $\frac{\partial Q(\hat{x}, \xi)}{\partial x_j}$ is integrable and if the residual satisfies $(1/h) \int_{\Xi \setminus N_\delta} \rho_j(\hat{x}, \xi; h) dP \xrightarrow{h \rightarrow 0} 0$, then $\frac{\partial Q(\hat{x})}{\partial x_j}$ exists and

$$\frac{\partial Q(\hat{x})}{\partial x_j} = \int_{\Xi} \frac{\partial Q(\hat{x}, \xi)}{\partial x_j} dP.$$

Differentiability of the recourse: discrete case

The residual condition $(1/h) \int_{\Xi \setminus N_\delta} \rho_j(\hat{x}, \xi; h) dP \xrightarrow{h \rightarrow 0} 0$ **cannot hold everywhere** here.

Indeed, in the linear framework with fixed recourse, and for vectors ξ with finite second-order moments, we have seen that $\mathcal{Q}(x)$ is piecewise linear when ξ has a finite support. It is therefore differentiable **except** at its finitely many kinks — and it is precisely at a kink that the residual condition fails. There, we will have to work with subgradients.

Differentiability of the recourse: continuous case

If ξ is continuous, $Q(x)$ is obtained as an integral over the $Q(x, \xi)$'s, that are not differentiable as they are piecewise linear given ξ . However, it is x that has to be fixed, not ξ . It is possible to show that (the proof is quite technical)

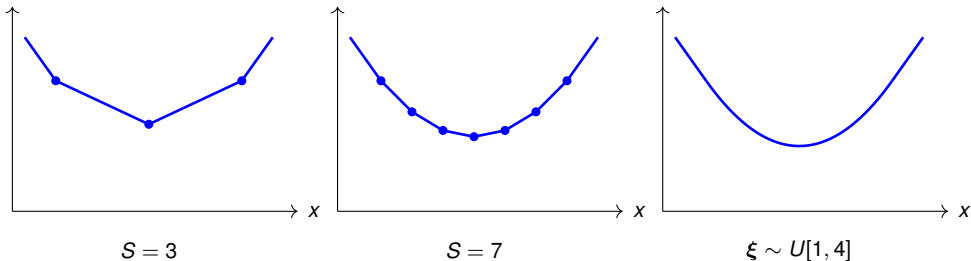
Theorem

For a stochastic program with fixed recourse where ξ has finite second-order moments, if ξ has an absolutely continuous distribution, then $Q(x)$ is differentiable on the interior of K_2 .

Intuitively, the function $Q(x)$ is “smoothed” by the superposition of an infinite number of functions $Q(x, \xi)$.

Smoothing by superposition

Same family $Q(x, \xi) = |x - \xi|$, with ξ spread more and more finely over $[1, 4]$.



The tails are identical; only the middle changes. With finitely many scenarios $\mathcal{Q}$ keeps one kink per scenario; in the limit the kinks are averaged away and $\mathcal{Q}$ becomes differentiable on the interior of K_2 .

The value of information

Uncertainty has a cost. Two natural questions:

- how much would we gain if the uncertainty could be removed **before** deciding?
- how much do we lose if we simply replace ξ by its mean?

Write the total cost attached to a first-stage decision x and a realization ξ as

$$z(x, \xi) \stackrel{\text{def}}{=} c^T x + Q(x, \xi).$$

The problem studied so far is the **recourse problem**

$$\text{RP} \stackrel{\text{def}}{=} \min_{x \in K_1 \cap K_2} \mathbb{E}_{\xi}[z(x, \xi)] = \min_{x \in K_1 \cap K_2} \{c^T x + Q(x)\},$$

and we denote by x^* one of its optimal solutions.

Wait-and-see

If ξ could be observed **before** deciding on x , we would solve one deterministic program per realization, and the expected cost would be the **wait-and-see** value

$$\text{WS} \stackrel{\text{def}}{=} \mathbb{E}_{\xi} \left[\min_{x \in K_1 \cap K_2(\xi)} z(x, \xi) \right].$$

Note the position of the expectation: in RP, a **single** x has to serve all the realizations, while in WS the decision may depend on ξ .

The **expected value of perfect information** is

$$\text{EVPI} \stackrel{\text{def}}{=} \text{RP} - \text{WS}.$$

It measures the maximum price a decision maker should be ready to pay for a perfect forecast of ξ .

$$\text{EVPI} \geq 0$$

Theorem

$\text{WS} \leq \text{RP}$, and therefore $\text{EVPI} \geq 0$.

Proof.

Let x be any feasible first-stage decision, i.e. $x \in K_1 \cap K_2$. For almost every realization ξ , x is feasible for the second stage, so

$$\min_{x' \in K_1 \cap K_2(\xi)} z(x', \xi) \leq z(x, \xi).$$

Taking the expectation preserves the inequality:

$$\text{WS} \leq \mathbb{E}_{\xi}[z(x, \xi)].$$

The left-hand side does not depend on x : minimizing the right-hand side over all feasible x gives $\text{WS} \leq \text{RP}$.

The expected value problem

A cheaper alternative is to ignore the randomness and to solve a single deterministic program built on $\bar{\xi} \stackrel{\text{def}}{=} \mathbb{E}[\xi]$: the **expected value (EV) problem**

$$\text{EV} \stackrel{\text{def}}{=} \min_{x \in K_1 \cap K_2(\bar{\xi})} z(x, \bar{\xi}),$$

whose optimal solution is denoted $\bar{x}(\bar{\xi})$.

$\bar{x}(\bar{\xi})$ is a legitimate first-stage decision, but it has been computed while ignoring the variability of ξ . Its true expected cost is the **expected result of using the EV solution**

$$\text{EEV} \stackrel{\text{def}}{=} \mathbb{E}_{\xi} [z(\bar{x}(\bar{\xi}), \xi)],$$

and the **value of the stochastic solution** is

$$\text{VSS} \stackrel{\text{def}}{=} \text{EEV} - \text{RP}.$$

$$\text{VSS} \geq 0$$

Theorem

$\text{RP} \leq \text{EEV}$, and therefore $\text{VSS} \geq 0$.

Proof.

By construction $\bar{x}(\bar{\xi}) \in K_1$; if it does not belong to K_2 , then $Q(\bar{x}(\bar{\xi})) = +\infty$, so $\text{EEV} = +\infty$ and the result is immediate. Otherwise, $\bar{x}(\bar{\xi})$ is a particular feasible point of the recourse problem, so

$$\text{RP} = \min_{x \in K_1 \cap K_2} \mathbb{E}_{\xi} [z(x, \xi)] \leq \mathbb{E}_{\xi} [z(\bar{x}(\bar{\xi}), \xi)] = \text{EEV}.$$



VSS measures what is gained by modelling the uncertainty explicitly, rather than replacing it by its mean.

A chain of bounds

Theorem

Assume that W , T and q are fixed, and that only the right-hand side is random, *affine* in ξ : $h(\xi) = h_0 + H\xi$ for a fixed vector h_0 and a fixed matrix H (e.g. $h(\xi) = \xi$). Then $\text{EV} \leq \text{WS}$.

Proof.

Set $\psi(\xi) \stackrel{\text{def}}{=} \min_{x \in K_1 \cap K_2(\xi)} z(x, \xi)$. Minimizing over x and y together, $\psi(\xi) = \varphi(h(\xi))$, where

$$\varphi(r) \stackrel{\text{def}}{=} \min_{x \geq 0, y \geq 0} \{c^T x + q^T y \mid Ax = b, Tx + Wy = r\}$$

is the value function of a single linear program in its right-hand side, hence convex (LP notes): this is where W , T and q must be fixed. Then

$$\text{EV} = \varphi(h(\mathbb{E}[\xi])) = \varphi(\mathbb{E}[h(\xi)]) \leq \mathbb{E}_\xi [\varphi(h(\xi))] = \text{WS}.$$

The *equality* is where affinity is used, $h(\mathbb{E}[\xi]) = h_0 + H\mathbb{E}[\xi] = \mathbb{E}[h(\xi)]$, and the inequality is Jensen's (probability notes), for the random vector $h(\xi)$. □

Neither assumption can be dropped

Take $c = 0$, $0 \leq x \leq 1$, $\xi = \pm 1$ with probability $1/2$ each, and a single second-stage constraint:

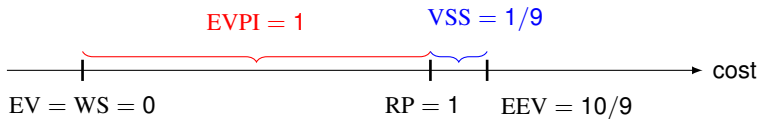
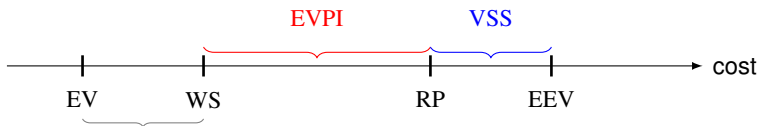
$$Q(x, \xi) = \min_{y \geq 0} \{y \mid y \geq h(\xi) - T(\xi)x\} = (h(\xi) - T(\xi)x)^+.$$

- **h random, not affine:** $h(\xi) = 1 - \xi^2$, $T = 0$. Then $\psi(\xi) = (1 - \xi^2)^+$, so $WS = 0 < 1 = \psi(0) = EV$. The red equality fails, $h(\mathbb{E}[\xi]) = 1 \neq 0 = \mathbb{E}[h(\xi)]$: the EV problem is built on the mean of ξ , not on the mean right-hand side.
- **T random:** $h = 1$, $T(\xi) = \xi$. Then $\psi(1) = \min_x (1 - x)^+ = 0$ and $\psi(-1) = \min_x (1 + x)^+ = 1$, so $WS = 1/2 < 1 = \psi(0) = EV$. Now ξ multiplies x : $Q(x, \cdot)$ is still convex in T for each fixed x , but the minimum over x is not — $\psi = 1$ on $[-1, 0]$ and $1 - \xi$ on $[0, 1]$.

Collecting the results: $WS \leq RP \leq EEV$ always holds, and when only $h(\xi)$ is random, affine in ξ ,

$$EV \leq WS \leq RP \leq EEV.$$

The four values on one axis



Example

Take again $Q(x, \xi) = |x - \xi|$ with ξ uniform over $\{1, 2, 4\}$, and assume that the first stage has no direct cost ($c = 0$), so that $z(x, \xi) = Q(x, \xi)$ and, as computed previously,

$$Q(x) = \begin{cases} 7/3 - x & \text{if } x \leq 1, \\ 5/3 - x/3 & \text{if } 1 \leq x \leq 2, \\ x/3 + 1/3 & \text{if } 2 \leq x \leq 4, \\ x - 7/3 & \text{if } x \geq 4. \end{cases}$$

- **RP**: the slope of Q changes sign at $x^* = 2$, the median of the scenarios, so $RP = Q(2) = 1$.
- **WS**: for each realization, $\min_x |x - \xi| = 0$, attained at $x = \xi$. Hence $WS = 0$, and $EVPI = 1$.

Example (cont'd)

- **EV**: $\bar{\xi} = \mathbb{E}[\xi] = (1 + 2 + 4)/3 = 7/3$, and $\min_x |x - 7/3| = 0$ at $\bar{x}(\bar{\xi}) = 7/3$, so $\text{EV} = 0$.
- **EEV**: the mean-value decision is now confronted with the three scenarios,

$$\text{EEV} = \mathcal{Q}(7/3) = \frac{1}{3} \left(\frac{4}{3} + \frac{1}{3} + \frac{5}{3} \right) = \frac{10}{9},$$

so that

$$\text{VSS} = \text{EEV} - \text{RP} = \frac{10}{9} - 1 = \frac{1}{9}.$$

We indeed have $\text{EV} = 0 \leq \text{WS} = 0 \leq \text{RP} = 1 \leq \text{EEV} = 10/9$. Here perfect information is worth much more than a better use of the distribution:

$$\text{EVPI} = 1 \gg \text{VSS} = 1/9.$$

Remarks

- There is no general ordering between EVPI and VSS: either one can be the larger.
- EEV, and therefore VSS, can be infinite: the mean-value decision $\bar{x}(\bar{\xi})$ may be infeasible for some scenarios. A relatively complete recourse rules this out.
- The definitions above are written for a minimization problem. For a maximization, $EVPI = WS - RP$ and $VSS = RP - EEV$. For the farmer problem of the first chapter, $WS = \$115406$, $RP = \$108390$ and $EEV = \$107240$, so $EVPI = \$7016$ and $VSS = \$1150$.
- Computing WS requires one deterministic program per realization, and EEV requires the recourse cost of $\bar{x}(\bar{\xi})$ over the whole support: both are out of reach as soon as the support is large, and are then estimated by sampling.

See Birge and Louveaux, Chapter 4.

Two-stage non-linear problems

Now consider the general program

$$\min_{x \in X} \mathbb{E}_{\xi}[f_0(x, \xi)] = \min_{x \in X} \mathbb{E}_{\xi}[g_0(x, \xi) + Q(x, \xi)].$$

Theorem

If $g_0(\cdot, \xi)$ and $Q(\cdot, \xi)$ are convex with respect to x , $\forall \xi \in \Xi$, and if X is a convex set, the aforementioned program is convex.

Proof.

For $x_1, x_2 \in X$, $\lambda \in (0, 1)$ and $x_{\lambda} = \lambda x_1 + (1 - \lambda)x_2$, convexity in x gives

$$\begin{aligned} g_0(x_{\lambda}, \xi) + Q(x_{\lambda}, \xi) \\ \leq \lambda(g_0(x_1, \xi) + Q(x_1, \xi)) + (1 - \lambda)(g_0(x_2, \xi) + Q(x_2, \xi)). \end{aligned}$$

Expectation preserves it, so the objective is convex; X being convex, so is the program.



In a more standard form

Inspired from Birge and Louveaux, Section 3.4.

We consider the problem

$$\begin{aligned} \inf z &= f^1(x) + Q(x), \\ \text{s.t. } g_i^1(x) &\leq 0, \quad i = 1, \dots, \bar{m}_1, \\ g_i^1(x) &= 0, \quad i = \bar{m}_1 + 1, \dots, m_1, \end{aligned}$$

where $Q(x) = \mathbb{E}_\omega[Q(x, \omega)]$ and

$$\begin{aligned} Q(x, \omega) &= \inf f^2(y(\omega), \omega), \\ \text{s.t. } t_i^2(x, \omega) + g_i^2(y(\omega), \omega) &\leq 0, \quad i = 1, \dots, \bar{m}_2, \\ t_i^2(x, \omega) + g_i^2(y(\omega), \omega) &= 0, \quad i = \bar{m}_2 + 1, \dots, m_2, \end{aligned}$$

We say that the recourse is additive (why?).

In a more standard form (cont'd)

The functions $f^2(\cdot, \omega)$, $t_i^2(\cdot, \omega)$, and $g_i^2(\cdot, \omega)$ are continuous for any given ω , and measurable w.r.t. ω for any given first argument. This allows one to prove that $Q(x, \omega)$ is measurable, and therefore that $Q(x)$ is well defined.

Reintroduce K_1 , $K_2(\omega)$ and K_2^s .

$$K_1 = \{x \mid g_i^1(x) \leq 0, i = 1, \dots, \bar{m}_1, \\ g_i^1(x) = 0, i = \bar{m}_1 + 1, \dots, m_1\},$$

$$K_2(\omega) = \{x \mid \exists y(\omega) \text{ s.t. } t_i^2(x, \omega) + g_i^2(y(\omega), \omega) \leq 0, i = 1, \dots, \bar{m}_2, \\ t_i^2(x, \omega) + g_i^2(y(\omega), \omega) = 0, i = \bar{m}_2 + 1, \dots, m_2\},$$

$$K_2^s = \{x \mid Q(x) < \infty\}.$$

Remarks

The formulation is not yet totally general. We will consider more general forms when we discuss sampling techniques.

Here, there is no longer a fixed recourse, but the first-stage decision x acts separately in the constraints. Goal: extend the previous results.

Questions: convexity, differentiability, optimality. We should also consider the concept of lower semi-continuity.